



Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

Document No.	DP-M4-20260708-DEVICEAL
Classification	Confidential - AI Draft
Generated	2026-07-08T07:33:22Z
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Firmware	v3.4
Framework	ISO 13485
Prepared By	DocPlus+ M4 Evidence and Documentation Layer

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

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This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

1. Document Profile and Controlled Metadata

Document Title	DocPlus+ Regulatory Complaint Investigation and CAPA Report
Document Number	DP-M4-20260708-DEVICEAL
Revision	A - AI Generated Draft
Complaint ID	CPL-20260708-7650
CAPA ID	CAPA-20260708-7643
RTM ID	RTM-DP-76507643
Classification	Confidential - AI Draft for quality and regulatory review
Generated Timestamp	2026-07-08T07:33:22Z
System of Record	DocPlus+ M4 Evidence and Documentation Layer
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Current Firmware	v3.4
Affected Component	alarm_manager
Regulatory Framework	ISO 13485

Data Quality and Source Provenance

This report references 50 graph fact(s): 13 controlled, 20 extracted, 9 inferred, 7 synthetic, 1 internal.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

Source Type	Referenced Fact Count
controlled	13
extracted	20
inferred	9
synthetic	7
internal	1

Digital-Twin Prototype Disclaimer

Hackathon prototype disclaimer: this AI-generated investigation may include digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

2. Regulatory Format Controls

ALCOA+ Data Integrity Mapping

Principle	DocPlus+ Implementation
Attributable	Prepared by DocPlus+ M4; generated timestamp and device scope are recorded.
Legible	A4 PDF with selectable text, structured headings, repeatable tables, and page numbers.
Contemporaneous	Generated at the time of complaint-report request from live agent state.
Original	Marked as AI Draft until Quality and Regulatory reviewers approve.
Accurate	Risk, CAPA, evidence, audit, and trace sections are populated from structured state.
Complete	Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields.
Consistent	Uses the same DocPlus+ controlled layout for all complaint-report exports.
Enduring	Saved as a PDF artifact under the application output directory.
Available	Downloadable through the M4 document endpoint and traceable by document number.

Applicable Regulatory Basis

Framework	Relevance to This Report
ISO 13485	Complaint handling, CAPA discipline, documented information, traceable quality records.
21 CFR Part 820	Design controls, complaint files, CAPA, and objective evidence expectations.
EU MDR	Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity.
21 CFR Part 11	Electronic record controls, auditability, attribution, and durable retrieval.

3. Executive Summary

Severity High	Risk Critical / RPN 20	Reportable True
Evidence Confidence medium (score 0.89)	Evidence Class Mix Candidate evidence - requires Quality review: 1, Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 1	Evidence Items 12
Audit Findings 2	Trace Decay Alerts 3	

Complaint: Device alarm always on and plays alarm everytime

Agent conclusion: Complaint DEV-PULSE-OX reports "Device alarm always on and plays alarm every time," severity High. The affected component is alarm_manager. Given the symptom and component match, the most plausible provisional cause is a software fault in alarm logic within alarm_manager (HYP-001). Prior, related alarm issues exist but concern missed/late alarms, not persistent alarms. Controlled verification runs for those prior issues on v3.4 passed and do not directly address an "always-on" scenario. With missing device identifiers (firmware, serial, lot) and no timeline, confidence remains moderate and provisional.

Risk Evidence Confidence Basis

Weighted evidence-class score 0.89 from 12 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 2. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-ALRM-052-V3-4: controlled_verification, RUN-TC-ALRM-053-V3-4: controlled_verification, RUN-TC-ALRM-054-V3-4: controlled_verification, RUN-TC-ALRM-057-V3-4: controlled_verification, RUN-TC-ALRM-058-V3-4: controlled_verification, RUN-TC-ALRM-059-V3-4: controlled_verification, RUN-TC-ALRM-051-V3-4: controlled_verification, RUN-TC-ALRM-055-V3-4: contro...

Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

4. Complaint Investigation Record

Field	Generated Value	Regulatory Basis
Complaint ID	CPL-20260708-7650	ISO 13485 8.2.2
Date Received	2026-07-08T07:33:22Z	21 CFR Part 11
Device UDI	UDI-DI-08717648-653846	EU MDR Article 27 / 21 CFR 830
Model / Version	DOCPLUS+-PULSE-OXIMETER / v3.4	IEC 62304 Section 8
Complaint Source	Post-market complaint intake	ISO 13485 8.2.2
Complaint Narrative	Device alarm always on and plays alarm everytime	21 CFR 820 complaint file
Patient Impact	Potential delayed awareness of a safety or status condition.	EU MDR Article 87
MDR Reportable?	Yes	EU MDR Article 87 / 21 CFR 803
CAPA Initiated?	Yes - CAPA-20260708-7643	21 CFR 820.100
Affected Component	alarm_manager	ISO 13485 8.5.2
Closure Status	Pending - Insufficient Evidence: Closure blocked: evidence class requires controlled verification before closure: CPL-100-055.	ISO 13485 8.2.2

Reportability Decision Logic

Condition	DocPlus+ Assessment	Action
Death or serious injury	Not stated in complaint text.	Monitor for escalation during intake and investigation.
Potential harm or near miss	Potential delayed awareness of a safety or status condition.	Treat as reportability review required.
Device relationship	Complaint maps to Alarm Notification.	Keep CAPA open until engineering evidence confirms or rules out relationship.
Malfunction/use issue	Reported condition indicates possible device malfunction or use-related failure.	Verify recurrence risk and effectiveness of controls.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CMP-005	3	Alarm sounded late during simulated desaturation. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-051	3	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	3	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-053	3	High pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-054	3	Low pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

5. Requirements Traceability Matrix

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-ALR M-051	High SpO2 alarm does not trigger.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-051	current and usable - Pass
REQ-ALR M-052	Low SpO2 alarm does not trigger.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-052	current and usable - Pass
REQ-ALR M-053	High pulse rate alarm missing.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-053	current and usable - Pass
REQ-ALR M-054	Low pulse rate alarm missing.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-054	current and usable - Pass
REQ-ALR M-055	Alarm sounds continuously without reason.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-055	current and usable - Pass
REQ-ALR M-056	Alarm volume cannot be adjusted.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-056	current and usable - Pass
REQ-ALR M-057	Alarm remains muted permanently.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-057	current and usable - Pass
REQ-ALR M-058	Alarm delay exceeds specification.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-058	current and usable - Pass
REQ-ALR M-059	Alarm icon does not appear.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-059	current and usable - Pass
REQ-ALR M-060	Alarm continues after condition is cleared.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-060	current and usable - Pass
REQ-SPO 2-003	Alarm behavior shall notify the user when saturation drops below configured thresholds. [DEMO DATA - NOT CONTROLLED EVIDENCE]	medium - M1 requirement scope [DEMO DATA - NOT CONTROLLED EVIDENCE]	Matched complaint terms against Design Control; relevance score 4.	missing - Candidate evidence - requires Quality review [DEMO DATA - NOT CONTROLLED EVIDENCE]	TEST-003, TEST-023	usable with caution - Pass, Needs Review

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-GEN-007	Post-market signals shall be linked to affected device models. [DEMO DATA - NOT CONTROLLED EVIDENCE]	medium - M1 requirement scope [DEMO DATA - NOT CONTROLLED EVIDENCE]	Matched complaint terms against Design Control; relevance score 3.	missing - Candidate evidence - requires Quality review [DEMO DATA - NOT CONTROLLED EVIDENCE]	TEST-015	usable with caution - Pass

Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

6. Design History File Index

#	Element	Document ID	Generated Unique Content	Status
1	Planning	DDP-DP-76507643	Design plan for complaint-impact review	Open
2	Design Input	URS-DP-76507643	Complaint-linked requirement review for Device alarm always on and plays alarm everytime	Draft
3	Design Output	SDD-DP-76507643	Alarm Notification design output references mapped	Draft
4	Design Review	DRM-DP-76507643	Cross-functional review required	Pending
5	Verification	VPR-DP-76507643	Objective evidence required for Alarm Notification on v3.4	Open
6	Validation	VAL-DP-76507643	Use-condition validation required for Device alarm always on and plays alarm everytime	Open
7	Design Transfer	DTR-DP-76507643	Release readiness blocked until evidence reconciled	Pending
8	Design Changes	DCR-DP-76507643	Change/service impact assessment linked to Alarm Notification	Open
9	DHF Index	DHFI-DP-76507643	DocPlus+ index for DocPlus+ Pulse Oximeter	Generated

Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving Alarm Notification. DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

7. EU MDR Technical Documentation - Annex II / III

Annex Ref	Section	DocPlus+ Generated Content
Annex II Section 1	Device Description	UDI UDI-DI-08717648-653846 scoped to complaint involving Alarm Notification.
Annex II Section 3	Design and Manufacturing Information	Design/change impact for Alarm Notification is carried into design output review.
Annex II Section 4	GSPR Compliance	Reported condition maps to safety/performance review: Device alarm always on and plays alarm everytime.
Annex II Section 5	Benefit-Risk Analysis	Critical risk complaint requires residual-risk review before closure.
Annex II Section 6.1	Verification and Validation	Objective verification evidence must be reconciled against the complaint scope.
Annex III Section 1.3	PMS / PSUR Input	Complaint is retained as a post-market signal for trend monitoring.
Annex III Section 1.5	Vigilance Reporting	Reportability remains open until human regulatory review confirms decision.

Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

8. Complaint Investigation Workflow and Execution Trace

8.1 Structured Complaint Intake Detail

Device ID	DEV-PULSE-OX
Affected Component	alarm_manager (COMP-ALARM-MANAGER)
Severity	High
Timeline	not specified
Symptom Codes	SYMPTOM_ALARM, SYMPTOM_DEVICE, SYMPTOM_ALWAYS, SYMPTOM_PLAYS, SYMPTOM_EVERYTIME
Match Terms	alarm
Similar Prior Complaints	5

Complaint Narrative

Device alarm always on and plays alarm everytime

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CMP-005	3	Alarm sounded late during simulated desaturation. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-051	3	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	3	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-053	3	High pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-054	3	Low pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

8.2 Investigation Chronology

Time	Agent	Action	Status	Detail
2026-07-08T07:28:10.493697Z	langgraph_supervisor	route to complaint_intake	info	Recorded
2026-07-08T07:28:10.494778Z	complaint_intake	extract structured complaint	started	Recorded
2026-07-08T07:29:22.738988Z	complaint_intake	extract structured complaint	completed	Elapsed 72244.5 ms
2026-07-08T07:29:22.739125Z	langgraph_supervisor	handoff check: similar incidents preserved for root cause	info	Structured event metadata recorded
2026-07-08T07:29:22.797794Z	langgraph_supervisor	route to root_cause	info	Recorded
2026-07-08T07:29:22.798893Z	root_cause	generate complaint-specific failure hypotheses	started	Recorded
2026-07-08T07:29:30.150746Z	root_cause	generate complaint-specific failure hypotheses	completed	Elapsed 7352.6 ms
2026-07-08T07:29:30.295998Z	langgraph_supervisor	route to firmware_traceability_ripple_check	info	Recorded
2026-07-08T07:29:30.296314Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	started	Recorded
2026-07-08T07:29:30.299335Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:29:34.193712Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:29:34.200396Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:29:34.200614Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	completed	Elapsed 3904.4 ms
2026-07-08T07:29:34.383580Z	langgraph_supervisor	route to evidence	info	Recorded
2026-07-08T07:29:34.383865Z	evidence	retrieve and validate evidence	started	Recorded
2026-07-08T07:29:40.199681Z	evidence	component-scoped evidence retrieval	info	Structured event metadata recorded
2026-07-08T07:29:40.200709Z	evidence	retrieve and validate evidence	completed	Elapsed 5816.8 ms
2026-07-08T07:29:40.347781Z	langgraph_supervisor	route to risk	info	Recorded

Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

Total wall-clock time	310529.89 ms
Agent steps	10
LLM calls	4
Tool/retrieval calls	25
Tokens	23091 total (16732 prompt / 6359 completion)
Longest agent	audit_shadow (135370.22 ms)

Agent Step Trace

Step ID	Agent	Action	Status	Elapsed	Sub-events
step-506cd9eba9	complaint_intake	extract structured complaint	success	72244.37 ms	LLM 1; tools 4
step-4232f1f386	root_cause	generate complaint-specific failure hypotheses	success	7352.06 ms	LLM 1; tools 2
step-ad414ba0c4	firmware_traceability_ripple_...	run firmware traceability ripple checks	success	3904.38 ms	LLM 0; tools 2
step-4cb8a285d8	evidence	retrieve and validate evidence	success	5816.98 ms	LLM 0; tools 3
step-f550a73545	risk	calculate ISO 14971 risk	success	406.73 ms	LLM 0; tools 2
step-953bd16be3	capa	draft CAPA with citations	success	4655.85 ms	LLM 0; tools 4
step-24b0f1c0f5	openai_reasoning	final complaint reasoning	success	18172.5 ms	LLM 1; tools 0
step-4668a727ea	audit_shadow	run adversarial audit	success	135370.22 ms	LLM 0; tools 5
step-3b92525eb7	openai_reasoning	final audit reasoning	success	14508.73 ms	LLM 1; tools 0
step-0794e15fd5	cybersecurity	scan SBOM components against NVD	success	48098.07 ms	LLM 0; tools 3

LLM Call Metadata

Call ID	Agent	Task	Model	Status	Tokens	Latency
llm-0b6a9268e2	complaint_intake	symptom_classification	gpt-5	success	1769 (1325/444)	4806.03 ms
llm-38ced26709	root_cause	root_cause_hypotheses	gpt-4o-mini	success	6006 (5250/756)	6256.48 ms
llm-34368d9c93	openai_reasoning	complaint_final_reasoning	gpt-5	success	7478 (4630/2848)	18165.73 ms
llm-5949ef2618	openai_reasoning	audit_final_reasoning	gpt-5	success	7838 (5527/2311)	14507.63 ms

Tool and Retrieval Call Metadata

Call ID	Agent	Tool	Query / Target	Results	Latency
tool-9bf94c31a7	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		117.64 ms
tool-2ab4641329	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		93.32 ms
tool-7ef7cc5e84	complaint_intake	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	63744.67 ms
tool-3ffb4b147	complaint_intake	graph_tools.find_similar_complaints	term=Device alarm always on and plays alarm everytime SYMPTOM_ALARM SYM...	5	135.12 ms
tool-9ec4927289	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-ALARM-MANAGER	417	960.25 ms
tool-21cee96cd8	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-ALARM-MANAGER	14	130.13 ms
tool-3b615bc275	firmware_traceability_rip...	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		149.73 ms
tool-558f39c605	firmware_traceability_rip...	graph_tools.run_ripple	device_id=DEV-PULSE-OX; new_version=v3.5; changed_components=['COMP-ALA...	10	3864.47 ms
tool-a280d9199a	evidence	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		125.73 ms
tool-4adba108e4	evidence	graph_tools.get_component_scoped_ev...	device_id=DEV-PULSE-OX; component_id=COMP-ALARM-MANAGER; component_name...	12	605.86 ms
tool-b22e245b03	evidence	vector_tools.retrieve_documents_debug	device_id=DEV-PULSE-OX; query=Device alarm always on and plays alarm ev...	8	5205.53 ms
tool-20987e6fe8	risk	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		83.56 ms
tool-cabf55a707	risk	graph_tools.open_capa_count	device_id=DEV-PULSE-OX	2	405.55 ms
tool-3c290d1c70	capa	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		110.86 ms
tool-9bf4f9e696	capa	graph_tools.get_audit_scope	device_id=DEV-PULSE-OX	120	1008.5 ms
tool-d9bb635dab	capa	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		104.7 ms
tool-d992478bbd	capa	graph_tools.get_capa_context	device_id=DEV-PULSE-OX	0	3633.51 ms
tool-292876ae5a	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		128.96 ms
tool-eb4dda46d7	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		132.5 ms
tool-c4d280a4a2	audit_shadow	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	79633.42 ms

5 additional tool/retrieval call(s) are retained in structured Trace AI data.

9. Root Cause Analysis

ID	Hypothesis	Probability	Affected Area	Key Why Chain
HYP-001	alarm_manager: Software Fault in Alarm Logic [DEMO DATA - NOT CONTROLLED EVIDENCE]	60%	COMP-ALARM-MANAGER	If the alarm conditions are improperly defined in the software, it can lead to continuous alerts. Historical issues in alarm systems frequently involve software misconfigurations. [DEMO DATA - NOT CONTROLLED EVIDENCE]
HYP-002	alarm_manager: Alarm Configuration Error [INFERRED DATA - REVIEW REQUIRED]	25%	COMP-ALARM-MANAGER	Configuration settings can be altered by users or through updates, potentially leading to erroneous alarm triggers. Similar incidents suggest that configuration issues have previously led to alarm malfunctions. [INFERRED DATA - REVIEW REQUIRED]
HYP-003	alarm_manager: Sensor Fault Leading to Continuous Alarm Trigger [DEMO DATA - NOT CONTROLLED EVIDENCE]	15%	COMP-ALARM-MANAGER	If the sensors reading for vital signs are inaccurate, it may continuously trigger alarms erroneously. Alarm systems are designed to react to sensor inputs, so faulty readings could lead to constant alarms. [DEMO DATA - NOT CONTROLLED EVIDENCE]

HYP-001 Evidence Balance

Evidence For	Device alarm always on and plays alarm every time. Affected component is identified as COMP-ALARM-MANAGER, which is responsible for alarm notifications. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Evidence Against / Gaps	No specific evidence confirming recent changes to the alarm logic or software. No recent software updates noted that could relate to this function. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Similar Incident Cross-Check	CMP-005: Alarm sounded late during simulated desaturation - suggests software timing issues but not directly the same problem. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Probability Rationale	The probability is high due to the software's critical role in alarm functions and previous similar issues within other alarm failures; however, no direct evidence confirms this is the root cause. [DEMO DATA - NOT CONTROLLED EVIDENCE]

HYP-002 Evidence Balance

Evidence For	Prior complaint (CPL-100-051 and CPL-100-052) identified issues with alarm trigger settings. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No direct evidence showing specific user error or configuration changes prior to the complaint. Not enough to conclude widespread configuration errors across models. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-052: Shows a pattern with alarm misconfigurations, suggesting user error or systemic issues. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	While configuration issues are plausible, the absence of specific incidents supporting widespread configuration errors reduces this probability. [INFERRED DATA - REVIEW REQUIRED]

HYP-003 Evidence Balance

Evidence For	User reported symptoms indicate that alarms trigger even without corresponding physiological conditions. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Evidence Against / Gaps	No evidence of defective sensor complaints reported before or concurrent that would indicate a pattern among devices. Current complaints are focused on the alarm system rather than the sensing elements. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Similar Incident Cross-Check	CMP-011 shows false readings but not specifically related to alarm management, indicating a potential correlation for future considerations. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Probability Rationale	This hypothesis is weaker compared to the first two due to minimal evidence suggesting a direct correlation between sensor faults and alarm triggers, despite it being a possible factor involved in alarm activations. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Root Cause Rationale

1. The reported symptom ("always on"/repeats every time) directly implicates alarm_manager; the intake tagged COMP-ALARM-MANAGER with matching terms, favoring HYP-001 (software fault in alarm logic).
2. Similar incidents (CPL-100-051/052/053/054) document alarm faults but of a different class (alarms not triggering or missing). Their stated root causes are diverse (threshold edge handling, inhibit timer not cleared, event queuing, unit mismatch) and were addressed with verification passing on v3.4; none explains a persistent/alarm-latched-always condition, so they neither confirm nor refute HYP-001 for this complaint.
3. Controlled verification evidence RUN-TC-ALRM-052-V3-4, RUN-TC-ALRM-053-V3-4, and RUN-TC-ALRM-054-V3-4 show Pass on v3.4 for the previously observed "missing alarm" failures. These do not provide reproduction or resolution data for an "always-on" alarm, leaving HYP-001 as a plausible but unproven explanation.
4. HYP-002 (configuration error) remains possible but lacks evidence of user/configuration changes; no specific configuration anomaly is cited in the complaint intake or evidence, lowering its likelihood relative to HYP-001.
5. HYP-003 (sensor fault driving false conditions) has limited support; no pattern of defective sensors is cited, and current complaints focus on alarm system behavior rather than sensing elements.
6. Key uncertainty: the complaint omits firmware_version, serial_number, lot, and timeline, preventing correlation with known fixed versions or environmental factors; therefore, the conclusion is provisional.

Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with Alarm Notification and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

10. Evidence and Provenance

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-ALRM-052 / TC-ALRM-052 / RUN-TC-ALRM-052-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-052: Low SpO2 alarm does not trigger. Test TC-ALRM-052: Verification test TC-ALRM-052; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-052-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-ALRM-052/TC-ALRM-052, Confidence: 0.96]
REQ-ALRM-053 / TC-ALRM-053 / RUN-TC-ALRM-053-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-053: High pulse rate alarm missing. Test TC-ALRM-053: Verification test TC-ALRM-053; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-053-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ve...	[Source: REQ-ALRM-053/TC-ALRM-053, Confidence: 0.96]
REQ-ALRM-054 / TC-ALRM-054 / RUN-TC-ALRM-054-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-054: Low pulse rate alarm missing. Test TC-ALRM-054: Verification test TC-ALRM-054; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-054-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ver...	[Source: REQ-ALRM-054/TC-ALRM-054, Confidence: 0.96]
REQ-ALRM-057 / TC-ALRM-057 / RUN-TC-ALRM-057-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-057: Alarm remains muted permanently. Test TC-ALRM-057: Verification test TC-ALRM-057; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-057-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-ALRM-057/TC-ALRM-057, Confidence: 0.96]
REQ-ALRM-058 / TC-ALRM-058 / RUN-TC-ALRM-058-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-058: Alarm delay exceeds specification. Test TC-ALRM-058: Verification test TC-ALRM-058; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-058-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3....	[Source: REQ-ALRM-058/TC-ALRM-058, Confidence: 0.96]

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-ALRM-059 / TC-ALRM-059 / RUN-TC-ALRM-059-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-059: Alarm icon does not appear. Test TC-ALRM-059: Verification test TC-ALRM-059; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-059-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verif...	[Source: REQ-ALRM-059/TC-ALRM-059, Confidence: 0.96]

10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

Hypothesis	Analysis Type	Result	Interpretation	Evidence Class
HYP-001	Firmware traceability ripple check	fail	HYP-001 evaluates alarm_manager: Software Fault in Alarm Logic. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hypot...	Firmware traceability signal - informational, not controlled verification evidence
HYP-002	Not applicable - alarm_timing_or_clinical_notification	not_run	HYP-002 evaluates alarm_manager: Alarm Configuration Error. The firmware traceability ripple check was not run because this failure mode maps to 'alarm_timing_or_clinical_notification', outside firmware-to-test traceability freshness review. This hypothesis...	Firmware traceability signal - informational, not controlled verification evidence
HYP-003	Not applicable - measurement_calibration_or_physiology	not_run	HYP-003 evaluates alarm_manager: Sensor Fault Leading to Continuous Alarm Trigger. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness...	Firmware traceability signal - informational, not controlled verification evidence

11. CAPA Report

Product	DocPlus+ Pulse Oximeter
Software Version	v3.4
CAPA Status	Pending - Insufficient Evidence
Priority	High
Risk Classification	Major - potential impact to Alarm Notification
Evidence Confidence	medium (score 0.89)
Evidence Class Mix	Candidate evidence - requires Quality review: 1, Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 1

Risk Evidence Confidence Basis

Weighted evidence-class score 0.89 from 12 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 2. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Insufficient Evidence Gate

This CAPA remains open for insufficient evidence. Closure blocked: evidence class requires controlled verification before closure: CPL-100-055. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure.

Problem Statement

A high priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: Device alarm always on and plays alarm everytime. In practical terms, the concern is that a user may not receive a safety or status notification at the expected time. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Complaint Summary

Reported condition: Device alarm always on and plays alarm everytime. Affected area: Alarm Notification. Reported timing or trigger: Not fully specified by complaint text. Observed symptom signals: alarm or notification behavior. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Immediate Containment Actions

Preserve the complaint evidence for Alarm Notification, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Investigation

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: Alarm Notification. For this complaint the scoped area is Alarm Notification, with emphasis on alarm or notification timing, event logs, and risk-control behavior. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 3 verification records need Quality review. Some verification evidence still needs Quality review before CAPA closure. Risk evidence basis: Weighted evidence-class score 0.89 from 12 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 2. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Root Cause Analysis

The current root-cause hypothesis is: A failure in the alarm management software that causes the alarm to continuously trigger, regardless of the actual condition of the device. This is a probable cause, not a final confirmed cause, because the CAPA record still needs returned-unit review, complaint evidence, trend review, and approved verification evidence showing whether the suspected cause is truly responsible. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Why CAPA Remains Open

The CAPA status is open because the available information is enough to justify corrective action, but not enough to close the quality record. Closure requires missing decisions or evidence: confirm or rule out the root cause, complete the correction or documented non-reproducibility rationale, verify notification timing and user-visible response with approved testing, and obtain Quality/Regulatory approval of the residual risk decision. Risk evidence confidence is medium (score 0.89); Medium uncertainty: 10 of 12 evidence item(s) are current controlled verification, but 2 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. This CAPA remains open because the scoped evidence chain is insufficient for closure; Closure blocked: evidence class requires controlled verification before closure: CPL-100-055. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Risk Assessment

The main hazard is that a user may not receive a safety or status notification at the expected time. The current risk level is Critical and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is medium (score 0.89); Medium uncertainty: 10 of 12 evidence item(s) are current controlled verification, but 2 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Corrective Actions

Reproduce or simulate the reported condition for Alarm Notification under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Rerun targeted verification, attach approved objective evidence, and resolve evidence gaps before release or closure. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Preventive Actions

Update the design, service, or release checklist so future changes affecting Alarm Notification include complaint-specific verification for alarm or notification behavior. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Verification of Effectiveness

Effectiveness must be verified by successful targeted testing of notification timing and user-visible response, review of complaint or service trends, and independent Quality review confirming that the affected area (Alarm Notification) meets approved design and risk-control requirements. The effectiveness package must replace candidate, historical, synthetic, or no-evidence gaps with approved objective evidence. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Effectiveness Criteria

The CAPA is effective when the reported condition no longer occurs or is formally ruled out, the behavior for notification timing and user-visible response is verified against the approved specification, all targeted verification activities pass, sufficient objective evidence is attached, and no related complaint recurs during six months after the corrected release or field correction. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Lessons Learned

Future complaint handling for Alarm Notification should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Closure

Closure decision: keep the CAPA open for insufficient evidence. The issue may warrant corrective action, but the scoped evidence chain still contains evidence that cannot support closure. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure. [INFERRED DATA - REVIEW REQUIRED]

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

11.1 CAPA Action Plan

CAPA Record	CAPA-20260708-7643
Complaint Reference	CPL-20260708-7650
CAPA Type	Corrective and preventive action
Current Status	Pending - Insufficient Evidence
Insufficient Evidence Gate	This CAPA remains open for insufficient evidence. Closure blocked: evidence class requires controlled verification before closure: CPL-100-055. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure.

Action Plan

Purpose	Owner	Required Action	How Closure Is Proven
Immediate measures	Quality / Support	Preserve the complaint evidence for Alarm Notification, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.	Evidence is preserved, affected users are protected, and deployment risk is understood.
Fix the issue	Software/System Engineering	Reproduce or simulate the reported condition for Alarm Notification under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Rerun targeted verification, attach approved objective evidence, and resolve evidence gaps before release or closure.	The confirmed cause is fixed or ruled out with approved objective evidence.
Prevent recurrence	Engineering / Quality	Update the design, service, or release checklist so future changes affecting Alarm Notification include complaint-specific verification for alarm or notification behavior. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.	Future changes or service decisions include controls for this failure mode before approval.
Check effectiveness	Quality Assurance	Effectiveness must be verified by successful targeted testing of notification timing and user-visible response, review of complaint or service trends, and independent Quality review confirming that the affected area (Alarm Notification) meets approved design and risk-control requirements. The effectiveness package must replace candidate, historical, synthetic, or no-evidence gaps with approved objective evidence.	No repeat issue is seen after targeted verification and evidence gaps are closed.

Acceptance Criteria for Closure

Closure Gate	Required Evidence
Reported condition	The reported condition is reproduced and corrected, or formally ruled out with approved evidence.
Affected function	Notification timing and user-visible response is verified against the approved requirement or acceptance criterion.
Verification	Missing, candidate, historical, synthetic, or no-evidence items are replaced by approved objective evidence.
Complaint monitoring	No related recurrence is observed during the defined follow-up period.
Approval	Quality and Regulatory reviewers approve the completed CAPA record.

12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

SBOM components	35
Components queried	25
NVD findings	7
Query date	2026-07-04
Source type	extracted
Cache status	cache_hit

Executive Severity Rollup

Critical	High	Medium	Low	None
0	2	5	0	0

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

SBOM Component Coverage

Component	Version	Supplier	CPE Candidate	NVD Result
Bootloader	1.8.0	Device OEM / Secure Boot Team	cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Embedded Firmware	2.4.3	Device OEM / Embedded Software Team	cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Operating System / RTOS	10.5.1	Amazon Web Services, Inc.	cpe:2.3:o:amazon:freertos:10.5.1:****:*	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
Device Drivers	1.6.2	Device OEM / Driver Integration Team	cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Signal Processing Software	3.2.1	Device OEM / Clinical Algorithm Supplier	cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Measurement Control Logic	2.1.0	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Alarm Management Module	2.0.5	Device OEM / Safety Software Team	cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
User Interface Module	1.7.4	Device OEM / UI Software Team	cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Settings Configuration Manager	1.9.1	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Data Storage Manager	1.5.0	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Patient Record Manager	1.3.2	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Communication Stack	2.2.0	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Bluetooth Stack	5.3.0	Nordic Semiconductor ASA	cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Wi-Fi Stack	2.10	w1.fi project / Device OEM Integration	cpe:2.3:a:w1.fi:wpa_supplicant:2.10:****:	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
TCP/IP Stack	2.1.3	lwIP project	cpe:2.3:a:lwip_project:lwip:2.1.3:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
USB Interface Software	1.4.1	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
File System	3.8.1	littlefs project / Device OEM Integration	cpe:2.3:a:littlefs_project:littlefs:3.8.1:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Logging Module	1.2.8	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Error Handling Module	1.1.6	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Time Management Module	1.0.9	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Self-Test Module	1.3.5	Device OEM / Test Software Team	cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Firmware Update Manager	2.0.2	Device OEM / Release Engineering Team	cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Authentication Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Authorization Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Cryptographic Library	3.0.13	OpenSSL Software Foundation	cpe:2.3:a:openssl:openssl:3.0.13:****:*	open_cves_found: 5 CVE record(s) returned by NVD for this CPE.

NVD CVE Findings - Structured Vulnerability Records

Vulnerability Record - CVE-2024-28115

CVE ID / NVD Reference	CVE-2024-28115 https://nvd.nist.gov/vuln/detail/CVE-2024-28115
CVSS Score / Severity / Vector	HIGH / CVSS 8.8 / v3.1 Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H
CWE ID(s)	CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification
Affected Component + Version	Operating System / RTOS 10.5.1
Vulnerability Description	NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact.
Exploitability / Exposure Context	Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks.
Recommended Controls	review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2024-28115

Vulnerability Record - CVE-2023-52160

CVE ID / NVD Reference	CVE-2023-52160 https://nvd.nist.gov/vuln/detail/CVE-2023-52160
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Wi-Fi Stack 2.10
Vulnerability Description	NVD describes an authentication or access-control bypass condition in the affected component.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2023-52160

Vulnerability Record - CVE-2009-1390

CVE ID / NVD Reference	CVE-2009-1390 https://nvd.nist.gov/vuln/detail/CVE-2009-1390
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-1390

Vulnerability Record - CVE-2009-3765

CVE ID / NVD Reference	CVE-2009-3765 https://nvd.nist.gov/vuln/detail/CVE-2009-3765
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3765

Vulnerability Record - CVE-2009-3766

CVE ID / NVD Reference	CVE-2009-3766 https://nvd.nist.gov/vuln/detail/CVE-2009-3766
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a certificate or TLS validation weakness in the affected secure-communication path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3766

Vulnerability Record - CVE-2009-3767

CVE ID / NVD Reference	CVE-2009-3767 https://nvd.nist.gov/vuln/detail/CVE-2009-3767
CVSS Score / Severity / Vector	MEDIUM / CVSS 4.3 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N
CWE ID(s)	CWE-295: Improper Certificate Validation
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3767

Vulnerability Record - CVE-2019-0190

CVE ID / NVD Reference	CVE-2019-0190 https://nvd.nist.gov/vuln/detail/CVE-2019-0190
CVSS Score / Severity / Vector	HIGH / CVSS 7.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H
CWE ID(s)	NVD-CWE-noinfo: No CWE information
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2019-0190

13. Electronic Records - 21 CFR Part 11

Requirement	DocPlus+ Implementation	Part 11
System Validation	DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks.	11.10(a)
Audit Trail	Generation record links CPL-20260708-7650, CAPA-20260708-7643, timestamp, and agent trace events.	11.10(e)
Access Controls	AI output is marked draft and requires QE/RA approval before controlled use.	11.10(d)
Electronic Signatures	Signature block captures reviewer, approver, date, and meaning of signature.	11.50
Record Integrity	PDF is generated as a stable artifact with a deterministic document number and retrievable filename.	11.10(c)
Human-Readable Copy	Selectable-text PDF is available through the document download endpoint.	11.10(b)
Retention	Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository.	11.10(c)

Generation Log Summary

Generation Hash	76507643
Input Fingerprint	7650764301C0995D
AI Draft Status	Pending human quality and regulatory review
Low Confidence Handling	Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values.

14. AuditShadow and Trace Decay

AuditShadow findings

Finding(s)	Requirement(s)	Risk	Observation	Remediation
FINDING-008, FINDING-020	REQ-GEN-007, REQ-SPO2-003	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Trace Decay alerts

Requirement(s)	Test(s)	Tested FW	Required FW	Reason
REQ-SPO2-003, REQ-GEN-007	TEST-003, TEST-023, TEST-015	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 12 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

Signal	What It Means	Source	Corrective Action
AuditShadow graph evidence gap	2 requirement finding(s) lack a VERIFIED_BY test link in M1.	M1 readiness score and traceability matrix	Create or link verification test evidence to each requirement and rescore readiness.
M4 retrieved evidence	12 candidate item(s) exist, but are not automatically controlled objective evidence.	M2 complaint pipeline and M4 evidence retrieval	Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware.
Trace Decay	3 firmware freshness alert(s) require review before release or approval.	M2 Trace Decay over M1 firmware/test links	Rerun verification on current firmware or document an approved equivalence rationale.
CAPA linkage	Open CAPA or complaint risk can keep readiness from being clean even when evidence exists.	M1 graph relationships and M2 risk assessment	Close, justify, or explicitly reference CAPA impact before final regulatory approval.

Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

14.2 AuditShadow Detail Appendix

Finding(s)	Requirement(s)	Reference	Risk	Observation	Remediation
FINDING-008, FINDING-020	REQ-GEN-007, REQ-SPO2-003	EU MDR PMS, IEC 60601-1-8	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

14.3 Trace Decay Detail Appendix

Requirement(s)	Test(s)	Tested Firmware	Required Firmware	Reason
REQ-SPO2-003, REQ-GEN-007	TEST-003, TEST-023, TEST-015	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

15. Cross-Standard Rendering Matrix

Element	FDA / QMSR	EU MDR	ISO 13485
Safety Requirement	Design input under QMSR; linked to verification evidence.	GSPR and Annex I safety/performance claim.	ISO 13485 Clause 7.3.3 design input.
Test Evidence	DHF verification record; current configuration/version required.	Annex II Section 6.1 V&V evidence.	Clause 7.3.6 design verification.
Complaint Record	Complaint file CPL-20260708-7650 with malfunction assessment.	Post-market vigilance signal under Articles 83/87.	Complaint handling under Clause 8.2.2.
CAPA Record	CAPA-20260708-7643 opened for Alarm Notification.	Trend/CAPA input for PMS and vigilance review.	Corrective action under Clause 8.5.2.
Risk Documentation	Risk analysis tied to design controls and complaint file.	Benefit-risk update in Annex II Section 5.	Risk management continuity through production/post-production.
Product Documentation	Lifecycle impact for Alarm Notification.	Design information and change-control evidence.	Validation and change-control evidence required.

Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

16. Document Control and Approval

Document No.	DOCPLUS-M4-20260708-76507643
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Revision Date	2026-07-08
Complaint ID	CPL-20260708-7650
CAPA ID	CAPA-20260708-7643
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Generated from	Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached
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Role	Name	Signature	Date
Quality Reviewer			
Regulatory Reviewer			